

$$\sum_{i=0}^m (s_i + 1)^2 = \sum_{i=0}^m (25i^2 + 20i + 4) = \sum_{i=0}^m 25i^2 + \sum_{i=0}^m 20i + \sum_{i=0}^m 4 = 25 \sum_{i=0}^m i^2 + 20 \sum_{i=0}^m i + 4 \sum_{i=0}^m 1$$

$$25 \cdot \frac{(m^2 - m) \cdot (3m + 1)}{6} + 20 \cdot \frac{(m^2 + m)}{2} + 4 \cdot (m + 1)$$

$$25 \cdot \frac{(2m^3 + m^2 + 2m^2 + m)}{6} \therefore 25 \cdot \frac{(2m^3 + 3m^2 + m)}{6} + 20 \cdot \frac{(m^2 + m)}{2 \cdot 3} + \frac{4m + 4}{6}$$

$$\frac{50m^3 + 75m^2 + 25m + 60m^2 + 60m + 24m + 24}{6} = \boxed{\frac{50m^3 + 135m^2 + 109m + 24}{6}}$$

Indução

$$P1) \frac{50 \cdot 0^3 + 135 \cdot 0^2 + 109 \cdot 0 + 24}{6} \therefore \frac{24}{6} = (4) \quad (5 \cdot 0 + 1)^2 = (4)$$

$$P2) S_m = S_{m-1} + a_m = \frac{50(m-1)^3 + 135(m-1)^2 + 109 \cdot (m-1) + 24}{6} + (5m+2)^2$$

$$\frac{50(m^3 - 3m^2 + 3m - 1) + 135(m^2 - 2m + 1) + 109m - 109 + 24}{6} + 25m^2 + 20m + 4$$

$$\frac{50m^3 - 150m^2 + 150m - 50 + 135m^2 - 270m + 135 + 109m - 85 + 250m^2 + 250m + 24}{6}$$

$$\boxed{S_m = \frac{50m^3 + 135m^2 + 109m + 24}{6}}$$

$$\begin{array}{r} 150 \\ 120 \\ +109 \\ \hline 379 \\ -270 \\ \hline 109 \end{array} \quad \begin{array}{r} 50 \\ +25 \\ \hline 135 \end{array}$$